

Homework 9, due Nov 5 before/in class

1. A particle is moving relativistically with Lorentz factor $\gamma \gg 1$ along a circle of radius R , Fig. 1. By aberration, a particle emits mostly within an angle $\sim 1/\gamma$ with respect to its instantaneous velocity. Estimate a distance Δx and corresponding time $\Delta x/c$ between first and last moment that the observer sees the particle. (An electromagnetic signal is first emitted at "1st observed wave emitted", and propagates with c . Particle nearly catches-up with its own radiation, so at the point "last observed wave emitted" it is behind the first emitted wave by Δx .)
2. Magneto-dipolar emission. A sphere of radius R carries dipolar magnetic field B_0 and rotates with spin frequency Ω . Estimate emitted power.
Hint: Check Landau & Lifshitz, vol 2, eq. (71.5)
3. In the problem above, the mass of the sphere is M (so that its moment of inertia is $I \approx (2/5)MR^2$). An observer can measure spin Ω and the rate of change $\dot{\Omega}$. Find magnetic field on the surface in terms of M , R , Ω and $\dot{\Omega}$.

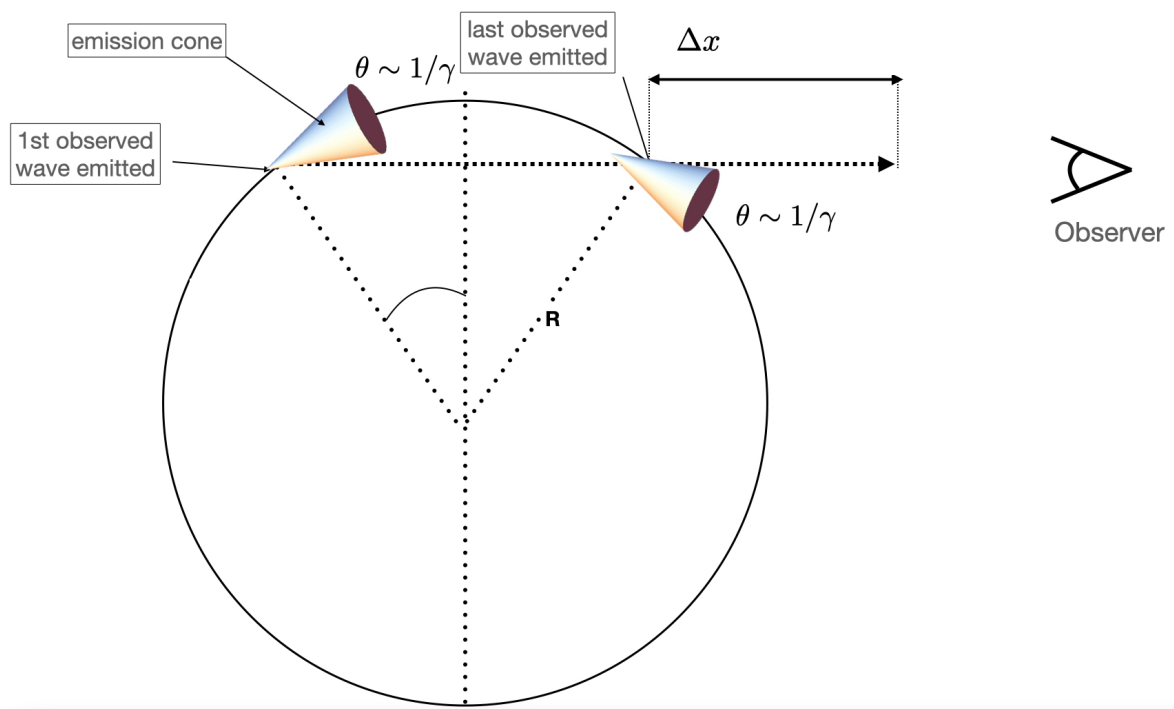


Fig. 1.—